

Independent prognostic factors for distant metastases and survival in patients with primary uveal melanoma

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Abstract

Adjuvant treatment strategies in uveal melanoma require determination of prognostic factors. Patients, who received primary therapy in 1994 and 1995 at our institution, were analysed. Of 271 patients 85% and 71% were available for follow up of ≥ 4 and ≥ 5 years. Forty three patients (15.9%) developed metastases. Kaplan–Meier analysis revealed a 5-year progression free survival (PFS) of 79% for the whole patient cohort. Extraocular tumor growth (EOG), ciliary body involvement or a largest tumor diameter (LTD) >14 mm were associated with a significantly lower 5-year PFS of 28%, 61.4% or 67.6%. In multivariate analysis time to progression was significantly associated with ciliary body involvement and LTD, and survival was associated with ciliary body involvement. Ciliary body involvement profoundly increased the risk for metastases (hazard ratio 6.9, $P<0.001$) within the first 3 years. This study determined patients with ciliary body involvement to be candidates for future adjuvant therapeutic interventions.

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1. Introduction

Uveal melanoma is the most common primary malignant tumor of the eye with an annual incidence of six cases in one million [1]. Studies in the early 1990s demonstrated that approximately 35% of patients will develop metastatic disease at various time points after primary therapy [2]. Metastases at the time

of first presentation are rare, indicating early hemato-genous spread [3]. Whereas primary uveal melanoma is subject to curative surgery or radiotherapy, only palliative treatment can be offered to patients with metastatic disease. Prognosis is poor with a median survival between 2 and 9 months after detection of metastases [4–6].

Clinical prognostic factors for metastases and survival have been identified in the 1970s and 1980s when enucleation was the standard therapy for uveal melanoma. These factors include ciliary body involvement, the largest tumor diameter (LTD) and the tumor height [7–9]. Since various studies demonstrated that

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